

task-1

1. Failing or Unseated RAM

Symptom: System powers on with fans spinning but produces a blank screen with repeating BIOS beep codes (e.g., continuous long beeps), or frequent random Blue Screen of Death (BSOD) crash loops.

2. Faulty Power Supply Unit (PSU)

Symptom: Total loss of power (no fan movement, no standby LEDs lit on the motherboard) or sudden, instant computer shutdowns/restarts without warning under heavy load.

3. CPU Overheating / Cooler Failure

Symptom: System turns on normally but abruptly powers off after 2 to 5 minutes of operation, often accompanied by maximum CPU fan speed noise and severe performance throttling before thermal shutdown.

4. Failing Storage Drive (HDD / SSD)

Symptom: "No Boot Device Found" or "Insert Boot Media" BIOS error upon power-on, clicking/grinding noises (mechanical HDD), or extreme system freezes during file read/write operations.

5. Defective or Unseated Graphics Card (GPU)

Symptom: Visual artifacts (streaks, flickering colored blocks, or screen tearing) on the display, display driver recovery crashes during 3D rendering, or a "No Signal" message on the monitor while PC power LEDs remain on.

Task-2

Laptop Not Charging – Step-by-Step Troubleshooting Plan

Step	What to Check	What to Do / Check
1. Check the power outlet		Verify that the wall Plug in another device, such as a phone charger or lamp. If it does socket is working. not work, try another socket.
2. Inspect the charger	Look for physical damage to the adapter, cable, or plug.	Check for cuts, bent pins, exposed wires, burn marks, or unusual smell. Do not use a damaged charger.

Step	What to Check	What to Do / Check
3. Check the charging connector	Examine the laptop's charging port.	Look for dust, dirt, bent pins, or a loose port. Make sure the connector fits firmly.
4. Check the charger connection	Ensure the adapter is properly connected.	Connect the charger firmly to the wall outlet and laptop. If the adapter has a separate power cable, check both connections.
5. Check charging indicators	Observe the laptop's LED.	Connect the charger and check whether the charging light turns on. No light may indicate a charger, port, battery, or motherboard problem.
6. Perform a power reset	Remove external power and reset the laptop.	Turn off the laptop, unplug the charger, remove the battery if it is removable, hold the power button for 15–30 seconds , then reconnect the charger and try again.
7. Check the battery	Inspect the battery if it is removable.	Remove and reinstall the battery securely. Check for swelling, leakage, or physical damage. Do not use a swollen battery.
8. Test with a known-good compatible charger	Determine whether the charger is faulty.	If available, use another charger with the correct voltage, connector, and required power rating . Never force an incompatible charger.
9. Check for overheating or burning	Inspect the laptop and charger for heat or smell.	If there is excessive heat, burning smell, smoke, or sparking, immediately unplug it and stop testing.
10. Check the charging port physically	Determine whether the port is loose or damaged.	Gently observe whether the connector moves unusually when plugged in. A loose port may require professional repair.
11. Use a multimeter if available	Test the charger output voltage.	Set the multimeter appropriately for DC voltage and compare the measured output with the voltage printed on the charger's label.
12. Identify the faulty component	Review the results.	If the outlet and charger are good but the laptop still does not charge, the likely problem is the charging port, battery, charging circuit, or motherboard . Professional repair may be required.

Task-3

Multimeter Test – USB Phone Charger

Objective:

To measure the output voltage of a standard USB phone charger using a multimeter and check whether it is within the expected USB range of **4.75 V to 5.25 V**.

Procedure

1. Plug the USB phone charger into a working electrical outlet.
2. Set the multimeter to **DC Voltage (V --)**.
3. Select a suitable voltage range, such as **20 V DC** if using a manual-range multimeter.
4. Insert the **black probe** into the USB connector's **GND (negative)** contact.
5. Touch the **red probe** to the **+5 V (positive/VBUS)** contact.
6. Read and record the voltage displayed on the multimeter.
7. Compare the measured value with the expected range **4.75 V–5.25 V**.

Observation

Test	Expected Voltage	Measured Voltage	Result
USB charger output	4.75–5.25 V	5.02 V	PASS

Conclusion

The measured output voltage is **5.02 V**, which falls within the expected USB range of **4.75 V to 5.25 V**. Therefore, the USB charger's output voltage is **normal**.

Task -4

Testing the ATX Power Supply 12V Rail

Steps

1. **Shut down the PC** and unplug the AC power cable.
2. Disconnect the **ATX power connector from the motherboard**. Do not test it while connected to the motherboard.
3. Set the multimeter to **DC Voltage (V --)**. Use a range such as **20 V DC** if it is not auto-ranging.
4. Identify the **yellow wire (+12 V)** and a **black wire (GND)** on the ATX connector.
5. Place the **red probe on the yellow wire**.

6. Place the **black probe on the black wire**.
7. Read the voltage displayed on the multimeter.
8. Record the reading and compare it with the expected **12 V rail**.

Observation

Test	Expected Voltage	Measured Voltage	Result
Yellow (+12 V) to Black (GND)	Approximately 12 V	11.2 V	Below 11.5 V – FAIL